SAAD ULLAH

Mathematics Graduate

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Personal Website | GitHub Profile

EDUCATION

Bachelor of Science in Mathematics 2021–2025

COMSATS University Islamabad, Islamabad Campus

Final-year project: *Elliptic Curve Cryptography*

Selected coursework: Abstract Algebra, Linear Algebra, Number Theory, Mathematical Analysis, Computational Mathematics, Probability and Statistics.

RESEARCH INTERESTS

Algebraic geometry, number theory, and the mathematical foundations of cryptography, particularly elliptic curves over finite fields.

RESEARCH AND INDEPENDENT STUDY

Elliptic Curve Cryptography 2025

Final-year project · COMSATS University Islamabad

Supervisor: Dr. Sohail Iqbal

- Studied elliptic curves over finite fields, their group structure, and the Elliptic Curve Discrete Logarithm Problem.
- Examined applications of elliptic curves to key exchange and digital signatures, including computational efficiency and security considerations.
- Prepared a final-year report connecting the underlying algebra and number theory with cryptographic protocols.

Independent Study in Algebraic Geometry

Undergraduate study · Faculty guidance: Dr. Sohail Iqbal

- Studied foundational ideas in algebraic geometry and their connections with elliptic curves and cryptography.

TEACHING EXPERIENCE

Mathematics Tutor Feb–Jun 2026

- Tutored students in Linear Algebra, Calculus, and Differential Equations through explanations, worked examples, and independent practice.

LaTeX Workshops — Lead Instructor and Organizer 2024 and 2025

Two editions · COMSATS University Islamabad

- Taught document structure, mathematical notation, tables, figures, bibliographies, theorem environments, and Beamer presentations.
- Prepared lecture slides and guided students through practical typesetting exercises.

Python Workshops — Lead Instructor and Organizer 2024 and 2025

Two editions · COMSATS University Islamabad

- Taught variables, expressions, control flow, functions, collections, and introductory Python libraries through examples and exercises.
- Prepared learning materials and made workshop resources available through my website.

ACADEMIC SERVICE AND LEADERSHIP

Graduate Study Preparation Circle — Organizer Oct 2025–Feb 2026

- Organized a peer learning group of approximately five to seven members preparing for graduate study.
- Coordinated shared discussions and admission-test practice; researched graduate programmes, entry requirements, and application procedures.

Emerging Trends in Mathematics, AI, and Computing Nov 2025

Organizer and Technical Coordinator · One-day symposium

SIAM CUI Chapter, COMSATS University Islamabad; organized with Middle Tennessee State University, USA.

- Supported technical arrangements and event operations for sessions on mathematics, artificial intelligence, and computing.

International Conference on Applied and Computational Mathematics 2025

Organizer and Technical Coordinator · COMSATS University Islamabad

- Supported speaker coordination, audio-visual arrangements, and parallel sessions at ICACM 2025.

Q K Ghori Alumni Seminar Series

Organizer and Technical Coordinator · COMSATS University Islamabad

- Helped organize the first five sessions, including speaker invitations, technical setup, audience coordination, and follow-up.

Tri-Math Quiz Competition — Event Organizer Nov 2024

COMSATS University Islamabad

- Coordinated faculty judges, team registration, and competition rounds covering speed mathematics, problem-solving, and theoretical questions.

HONORS AND AWARDS

Undergraduate Merit Scholarship 2022

COMSATS University Islamabad; awarded for academic performance.

First Runner-Up — Battle of Brain Quiz Competition

NUML Islamabad.

TECHNICAL SKILLS

Programming: Python; introductory C++.

Academic typesetting: LaTeX, Beamer, mathematical documents, bibliographies, and presentation slides.

Teaching resources: Lecture notes, worked examples, programming exercises, and workshop slides.

PROFESSIONAL AFFILIATION

Member, SIAM CUI Student Chapter, COMSATS University Islamabad.